

A Simple Approach to Tracking the Movement of a Robot Vehicle from Solely Optical Camera Input.

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2021-06-09

Abstract

The ability of an autonomous robot to accurately and reliably estimate its position and orientation is critical if it is to be able to plan and carry out future movements. It is preferable to combine different types of measurement in order to produce the best estimate. Odometry, through the measurement of motion, provides a valuable complement to absolute position measurement. Visual odometry uses an optical camera as a motion sensor, and through its different strengths and weaknesses, can be used to augment or replace other motion sensors like wheel encoders and inertial sensors. The large quantity and broad nature of the data provided by a camera sensor presents a challenge in how to efficiently obtain accurate motion measurements. There exist countless methods. This report describes a simple implementation of visual odometry using mobile phone cameras to record video input. It presents a method using the Kalman filter to match detected features over successive video frames and thus measure motion.

1 Introduction

Movement tracking is of crucial importance to autonomous robot vehicles. In the applied estimation course we have used the extended Kalman filter to combine relative pose estimates from wheel odometry with absolute pose estimates from range and bearing measurements to known landmarks. Having taken a computer vision course in parallel I was keen to try movement tracking using only video data as input. I also wanted to try out the ideas we had learned in a practical way with some simple hardware.

The problem then is how to track movement using only a camera, or cameras, as input. This is a problem that was first studied in the 1980s beginning with Moravec (1980) and has generated much interest with the increasing computational resources required to perform image processing in real time. The problem is interesting in part because of the ubiquitousness and relative cheapness of optical cameras compared to other forms of input. Cameras also provide broad data that are useful in multiple applications meaning they allow for great flexibility and potential for expansion into other uses. It also means there is a good chance a robot already has one. A camera is also a passive sensor unlike lidar and sonar and so does not generate and is not susceptible to interference from other robot vehicles using active sensors (Aqel et al. 2016).

Nistér, Naroditsky and Bergen (2004) demonstrated that camera input can provide so called visual odometry that is more robust and less prone to error accumulation (drift) than wheel odometry, especially on uneven terrain and where wheel slippage is a con-

cern. This has made it a valuable technique for rovers exploring the Martian surface, helping to improve vehicle safety and saving time for scientific observations by reducing the amount of human intervention needed during navigation. (Maimone, Cheng and Matthies 2007). As on other planets there are many places where the Earth's GPS satellites are obscured, such as indoors, underground, underwater, in canyons and among tall buildings, where camera based tracking can help.

The big advantage of wheel odometry is that it is cheap and simple. Camera based tracking can play a similar role in non-wheeled robot vehicles. It is much less prone to error accumulation than inertial sensors of comparable cost. For autonomous aerial vehicles, which already lend themselves well to a role as a camera platform, and where weight savings are critically important, it makes a lot of sense to also use the camera to track the movement. While aerial vehicles are the obvious example of non-wheeled robots, submersibles are another, and more excitingly, visual based tracking opens up possibilities for novel and unconventional robot designs like the vine robots demonstrated by Hawkes et al. (2017).

Visual based movement tracking can be divided into two approaches. The first is to estimate absolute pose through range and bearing measurements to known landmarks. The second is to estimate movement of the camera and thus the robot and so determine the new pose relative to the previous estimation. This second approach is what is known as visual odometry. These two approaches are complementary and can be married through the use of probabilistic tools like the Kalman filter.

With both approaches it is first necessary to detect features in the image that correspond to objects in the environment. For absolute pose estimation the features need to be matched to mapped landmarks and their range and bearing measured. For visual odometry any features can be used, as long as they are real and static in the environment. Range measurement is also needed in visual odometry in order to assess the scale of the movement.

Bearing measurements can be obtained straightforwardly through knowledge of the camera optics. The most obvious way to measure range is through stereoscopic vision, but it is also possible with a single camera when the actual size of an object is known, or when the motion of the camera is constrained in some way. For visual odometry it is also necessary to track how an object moves in the image over time. This requires a way of matching an object with itself as it appears in consecutive video frames. Finally all these data must be combined to estimate the robot's pose.

Here I use a stereo camera feed to detect objects standing up on an otherwise flat floor. The stereo input is however not used for depth perception and the approach used to calculate distance would also

work with a single camera. The object detection method is in fact quite crude and has many drawbacks. It was chosen as a quick and dirty way to generate measurements without focusing too much on computer vision. In retrospect, given the difficulties in calibration between the two cameras, any other feature detection method performed with one camera would have been better.

The initial idea was to use the objects as landmarks for range and bearing measurements and investigate ways to deal with the messy data using a Kalman filter. The first approach then was a zero odometry approach. It always assumed the robot was stationary and only updated its pose estimate through measurement of the landmarks. This approach was very weak and so visual odometry was introduced in a second approach.

Visual odometry was implemented through use of a second Kalman filter that estimated the robot's linear and rotational velocity. This estimate was used by the first Kalman filter to predict its new position (no longer assuming constant pose). The second Kalman filter was a derivative of the first. It assumed a constant forward and rotational velocity which it updated through measurement of the change between two consecutive video frames.

This method suggests obvious extension using a second derivative Kalman filter to estimate the robot's linear and rotational acceleration. It would use a measurement of change over three consecutive frames to update its estimate.

1.1 Contribution

This work describes the implementation of a simple visual odometry system using the Kalman filter to match features across a sequence of video frames.

2 Related Work

Moravec (1980) is credited as being the first to track the pose and movement of a robot purely from camera images (Nistér, Naroditsky and Bergen 2004; Aqel et al. 2016; Scaramuzza and Fraundorfer 2011). At a time when Stanford Research Institute and NASA Jet Propulsion Laboratory were experimenting with laser range finders for autonomous robot control Moravec was constrained by circumstance to a robot cart from the 1960s with a single TV camera. Moravec realised that while laser range finding has the advantage that it provides simple, specific data, a camera, by providing a broader range of data has great potential for wide ranging applications. A camera can also be used to estimate distance, albeit at a much greater computational cost. Moravec used both a slider to move the camera on the stationary robot and the movement of the robot itself to generate images from multiple perspectives and so construct a 3D model of the robot's environment. The process was slow and halting, taking five hours to navigate a 20 meter course.

Nistér, Naroditsky and Bergen (2004) coined the term visual odometry to refer to the use of cameras to track the relative motion of a robot. They demonstrated an algorithm fast enough to be performed online and their work forms the basis of many visual odometry approaches to date. Their method consists of

three basic elements which serve as a useful point of comparison for subsequent approaches. These are: *feature detection*, whereby features (such as corners) are located in the camera image; *feature matching*, where a feature as it appears in one image is matched with itself as it appears in other images (from the other camera in a stereo visual system and from previous frames); and *pose estimation* where the changes in the position of features from frame to frame are used to estimate the change in the robot pose. Nistér, Naroditsky and Bergen tracked features in the image plane, which was observed to yield far superior results than tracking them in 3D, due to the greater uncertainty in the calculation of the depth of features in the visual field.

The greatest challenge for visual odometry is scale uncertainty. This is the uncertainty in calculating the scale of the movement of features from one frame to the next, and depends on the uncertainty in how far those features are from the camera. Consequently the most successful earlier methods use stereo camera feeds. Stereo cameras allow for depth perception but require careful calibration. Their ability to perceive depth also diminishes with distance relative to the baseline distance between the two cameras. To accurately gauge the distance of far away features requires that the robot's eyes be further apart and so stereo visual systems are greatly constrained by the size of the robot. At a certain distance a stereo system becomes essentially mono.

Consequently, increasing attention has been devoted to monocular systems, which are both cheaper and lighter than stereo camera systems and more applicable to smaller robots. One such approach that is particularly relevant to this project is that of Van Hamme et al. (2015). The disadvantage with monocular systems is that they require some constraint on the camera's motion in order to perceive depth. Van Hamme et al. demonstrate an approach based on detecting features on the road surface around a vehicle. This means that the angle of the camera relative to the road surface is constrained to within a relatively small set of angles. Furthermore, features are constrained to move within the ground plane. Van Hamme et al. track feature movements in the ground plane instead of the image plane using a reverse perspective projection. To match features over successive frames they use a method similar to a Kalman filter, using a prior estimation of the vehicle's movement to predict the current location of previously observed features and then the measurement of those features in the current frame to update the estimate.

The method presented in section 3 shares these two elements of tracking features in the ground plane by using reverse perspective projection, and performing feature matching using prediction from the prior state estimate, although in this case using a straight-forward extended Kalman filter without the more sophisticated estimation methods that Van Hamme et al. employ. This simpler approach is possible because the robot operates in a simpler environment where the camera height and declination remain stable over a flat floor. Although not robust to more complex environments, I think this method presented here is at least conceptually interesting and lends itself to easy extension to deal with more complex movements.



Figure 1: Perspective mapping from the image plane (left) to the floor plane in the robot reference frame (right). The yellow earplugs used as landmarks can be seen running diagonally in the image from bottom right to top left. The right hand image shows how the effective resolution of the projected image diminishes with distance, leading to increasing measurement uncertainty. Only the points on the floor immediately in front of the robot and visible to both cameras are mapped from the source images. The quality of data deteriorates outside this area and the noise increases.

3 Method

3.1 Feature Detection

The feature detection method was originally intended purely as a way to detect mapped landmarks sticking up out of the otherwise flat floor. It was achieved by comparing the difference between perspective mappings of the two camera feeds. The idea is to transform the camera images in such a way so as to map points (X, Y) in the image plane onto their respective points (x', y') , on the floor in the robot frame of reference. Under this transformation the image is distorted to become a scale map of the floor plane. The result of this transformation is depicted in figure 1.

Any 3D objects in the image that are not in this plane appear distorted as though they were trompe l'œil painted on the floor. To detect these objects, the transformed image from one camera is subtracted from its simultaneous partner from the other camera. In the resulting image, where the cameras' fields of view overlap, the floor will be darkened as it appears similar to both cameras. Objects, which present a different side to each camera and obscure different areas of the floor behind them stand out as lighter areas. Gaussian smoothing was used to reduce noise and detail in this image. This reduces false detections by preferring larger features. A set of n brightest local maxima in the difference image formed the set of detected features. An overview of these steps is shown in figure 2.

3.2 Estimation Without Odometry

Estimation of the robot's pose was performed using an extended Kalman filter (EKF) with state vector $\mathbf{x} = (x, y, \theta)$ representing the robot's position and heading in the map. The first attempt did not use odometry and in the prediction step assumed no change in the mean state μ . The mean state only changed through measurement update. Detected features were converted to range

and relative bearing measurements $z = (r, \phi)$. Batch update was utilised, using maximum likelihood to associate measurements with known landmarks on the map and reject outliers.

The reason range and bearing were used in the observation model was to be better able to model measurement uncertainty. While uncertainty in the range measurement increases non-linearly with range, uncertainty in the bearing was believed to be constant. This is actually incorrect since the bearing is from the origin in the robot frame which is offset from the two camera frames.

3.3 Estimation With Visual Odometry

To improve the accuracy of the pose estimate, odometry was introduced as a control variable $u = (v_x, v_y, \omega)$, representing the robot's linear and angular velocity. The velocities were estimated using a second Kalman filter with state vector $\dot{\mathbf{x}} = (v, \omega)$, representing the robot's forward and angular velocity. This filter assumed no change in the mean velocity estimate during the prediction step, updating only according to how the range and bearing measurements of detected features changed from one video frame to the next. The velocity estimation EKF was essentially a clone of the pose estimation EKF, but with a different observation model.

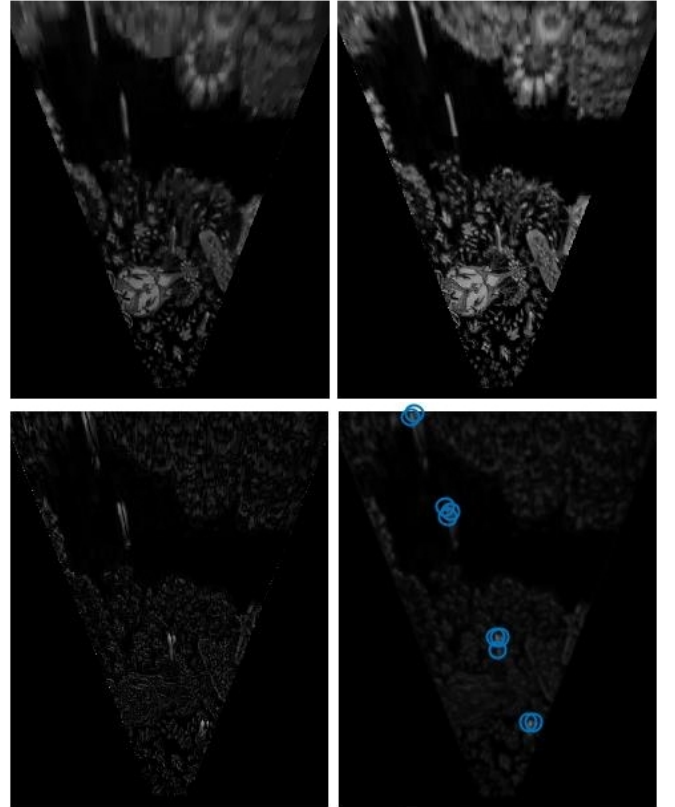


Figure 2: Feature detection for a single video frame. Top left and right show the perspective mappings from the left and right cameras. Bottom left depicts the absolute difference between these two projections, showing how the brightness of the rug pattern is reduced but not eliminated. Bottom right shows this image after Gaussian blurring and with the ten brightest local maxima circled. These are the detected features. In this example all four visible landmarks are correctly identified, including those in the noisy patterned area of the rug. There is a systematic bias caused by the strongest response coming from the top of each landmark.

3.4 Feature Matching

To measure how the detected features move relative to the camera relies on being able to match features with themselves across consecutive video frames. Nistér, Naroditsky and Bergen (2004) for example match features within a fixed separation from each other across two images, and use a mutual consistency check to only accept matches where both features prefer to match with each other. The method used here is similar to that used by Van Hamme et al. (2015). The estimation of the robot's movement is used to predict where a detected feature in one video frame will appear in the next. The same approach is used as with matching detected features to known landmarks, i.e. a batch update using maximum likelihood association and outlier detection. This was done using the observation model $\mathbf{h} = (r_k, \phi_k)$,

$$r_k(v, \omega) = \sqrt{r_{k-1}^2 + (v\Delta t)^2 - 2r_{k-1}v\Delta t \cos \phi_{k-1}}, \quad (1)$$

$$\phi_k(v, \omega) = \arcsin\left(\frac{r_{k-1} \sin \phi_{k-1}}{r_k}\right) - \omega\Delta t, \quad (2)$$

with Jacobian,

$$H_{11} = \frac{2v\Delta t^2 - 2r_{k-1}\Delta t \cos \phi_{k-1}}{r_k}, \quad (3)$$

$$H_{12} = 0, \quad (4)$$

$$H_{21} = \frac{-r_{k-1} \sin \phi}{r_k^2 \left(1 - \frac{r_{k-1} \sin \phi_{k-1}}{r_k}\right)^2} H_{11}, \quad (5)$$

$$H_{22} = -\Delta t, \quad (6)$$

where Δt is equal to the inverse of the video frame rate.

3.5 Camera calibration

For the feature detection described in section 3.1 to work, the projection transformations for each of the cameras needs to be accurate and fairly precise. Rough measurements were taken of the relative orientations of the two cameras with the floor, from which precision of the calibration was increased through an iterative process. Images of a flat section of floor with no raised features but with a patterned surface were used to perform the calibration. The calibration process used gradient descent to minimise the difference between the two images through small changes in the perspective projection parameters. These parameters described the offset in position and orientation between the cameras, the focal length of each camera, and a factor to correct for difference in exposure.

3.6 Implementation

A stereo video dataset was created using two mobile phone cameras supported upright in a box which was pushed by hand over the floor. Foam earplugs were used as features and laid out according to a map. The robot was moved along a rectangular path with the same start and finish point. Two different floor surfaces were used. A matte wooden floor with minimal patterning from

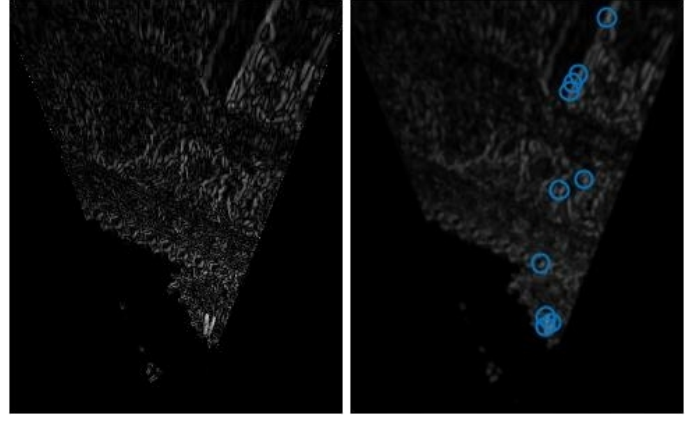


Figure 3: The left hand image depicts the absolute difference between left and right cameras, and the right hand image after blurring, with the ten brightest local maxima circled. In this example only one landmark was in the field of view. It is visible at the bottom of the image giving rise to the cluster of detections closest to the robot. This example illustrates how a large number of erroneous detections occur when there are too few visible landmarks. While these detections result in false measurements to the pose estimation EKF, any that are detected consistently from frame to frame will be useful measurements for visual odometry. Detections too far from an expected landmark are rejected as outliers for pose estimation. The velocity detection EKF only rejects those that are too far from where it expects to find detected features from the previous frame.

the wood grain and a colourful patterned rug. Both surfaces were chosen to have little light reflection. The patterning would be a test of the perspective transformation. In theory, with good camera calibration, any pattern in the floor should be eliminated when comparing the two perspective transformed images.

The two cameras were aligned as best as possible to be at the same height, pan, tilt, and roll, and to have the same focus and exposure settings. The videos were converted to a series of image frames using ffmpeg at a rate of 24 frames per second. Synchronisation of the videos was achieved by recording a camera flash and finding the frames over which the flash occurred. Since this was done after converting the video to images, synchronisation was accurate to within 1/24 s.

The original colour image frames were converted to greyscale by taking one colour channel. This was done rather than averaging the RGB values to eliminate the affect of any discrepancy in white balance between the two cameras. Projected images were then created by sampling from the greyscale images using the lookup table for the perspective projection. For sampling, both nearest neighbour and bilinear interpolation were tested. For the overall accuracy of the tracking algorithm the faster nearest neighbour method performed as well as bilinear interpolation and so was the method used for all analysis.

Difference in exposure between the two images was compensated for by linearly scaling the greyscale values of one of the images before calculating the difference image. Local maxima were identified as those pixels that were brighter than their 8 neighbours.

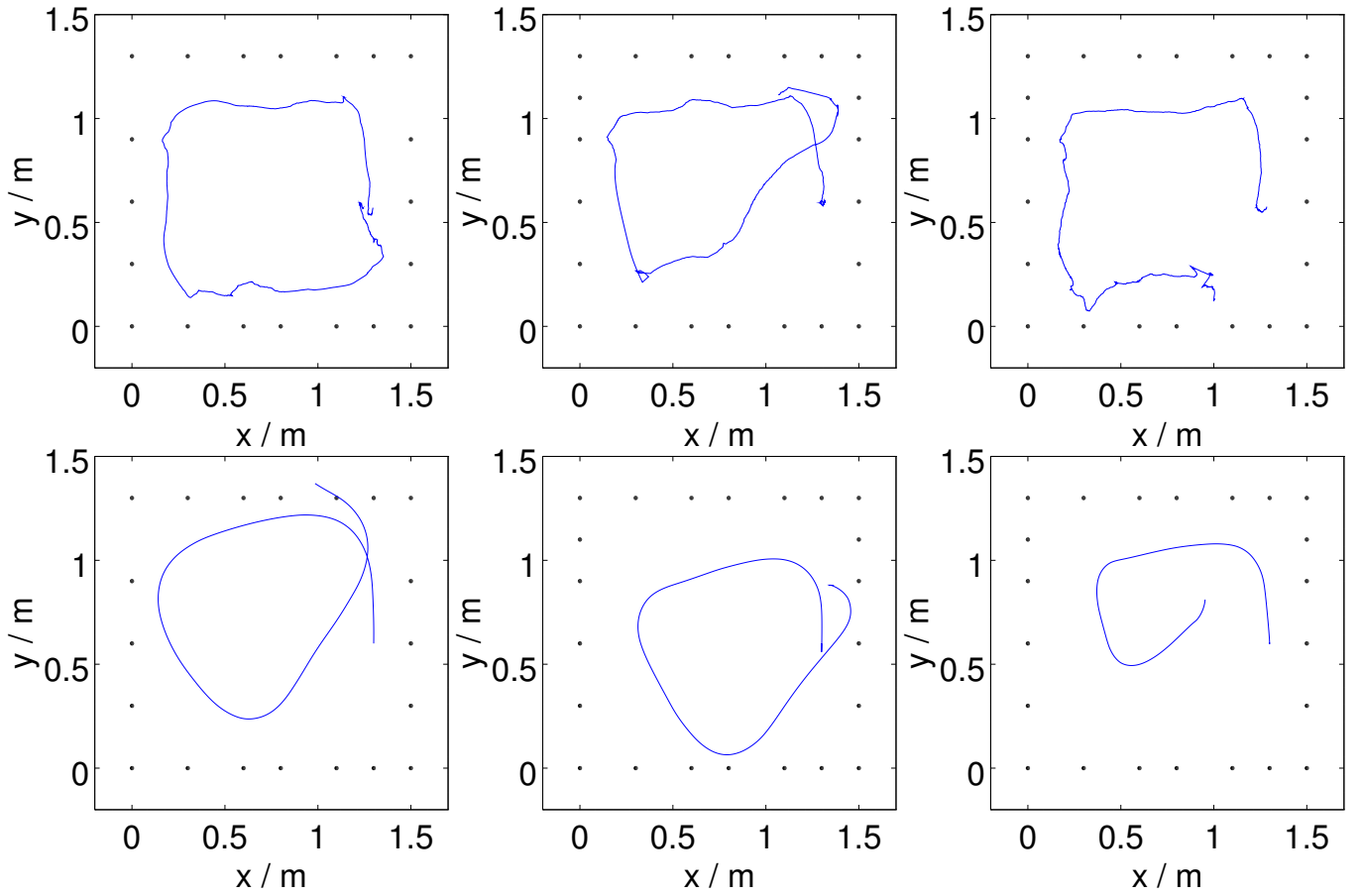


Figure 4: Tracking results from three sets of video data (left to right), corresponding to three separate runs around the course. The first two were performed on a patterned rug, the third on the bare wooden floor, which had a matte finish. The process and measurement noise covariance matrices were tuned to perform as well as possible on the first (leftmost) dataset, and are the same for all datasets. The top row shows tracking of the robot position by combining visual odometry from the velocity estimation EKF and an update of the position and heading from measurements to identified landmarks. The bottom row shows visual odometry alone. The landmarks are depicted by small circles and form a rectangle around the path of the robot. The robot started and finished at $(1.3, 0.6)$ facing in the positive y -direction. It moved anticlockwise on a closed rectangular path, making smooth curved turns (i.e. it did not turn on the spot). The kinks in the track after each turn show the pose estimation adjusting to the measurements from newly visible landmarks, and illustrate the effect of the biased measurements. Visual odometry can be seen to always overestimate the rotational velocity in the turns. Visual odometry also estimated a continuous slight left turn when the robot was moving straight ahead. It also appears to have a tendency to slightly overestimate forward velocity except in the third set where it is dramatically underestimated.

4 Results

It is difficult to do a quantitative evaluation of the performance of tracking method because there is no ground truth to compare the estimated states to. There are however a good number of things to be said qualitatively about its performance.

It was difficult to achieve good calibration of the two cameras and effectively reduce features in the plane of the floor such as the pattern of the rug and grain of the floorboards. A large amount of such features were present in the final image and gave rise to erroneous measurements, as can be seen in figure 3. The measurements were therefore often very noisy with frequently a large number of measurements that needed to be rejected as outliers. On the plus side the pose estimation observation model was generally always able to identify a landmark when it was in the frame.

There was also systematic bias in the range detection of landmarks. The method used to determine the range and bearing to a landmark was quite crude with all areas of the floor obscured

by the landmark potentially giving rise to a measurement. Landmarks were therefore systematically measured as being further away than they actually were leading to a systematic error in the estimate of the robot's position. This was especially visible when the robot was cornering as the error, which was along the robot's line of sight before the turn became an error across the field of vision after. Consequently the observation model would expect landmarks at an offset bearing. Without visual odometry, in the worst cases, the error resulted in the robot failing to correctly assign the new landmarks and a complete failure to track its position and orientation.

The addition of visual odometry helped greatly in improving tracking performance. Without odometry it was necessary for the Kalman filter tracking (x, y, θ) to model process noise in the order of a centimeter for it to be responsive enough to the measurements. This led to erratic tracking of the robot's position and heading. The addition of the visual odometry allowed for process noise to be modelled at the order of a millimeter and had a stabilising effect on the position tracking. The performance of the tracker on three separate runs can be seen in figure 4.

Accumulation error in the visual odometry was significant and compares poorly to previous work, for example (Nistér, Naroditsky and Bergen 2004). The visual odometry system consistently overestimated its rotational velocity when cornering and estimated a continuous slight turn away from the visible landmarks when it was actually moving straight forward. It also slightly overestimated its forward velocity when moving in a straight line. This can be explained by the bias in the measurement to the landmarks, which are the strongest features detected when they are in view. The velocity estimation EKF only considers the relative movement of these features so the bias presents differently than with pose estimation. The landmarks appear to the feature detection system in the shape of their shadows as cast from two light sources, one at the focal point of each camera. As the robot moves towards a landmark these 'shadows' become shorter and the range bias decreases. This will make the landmark appear to approach faster than in reality and result in an overestimation of the forward velocity.

It might be expected that the bearing measurement would be unaffected by this bias. However, due to the two camera setup, a landmark over to one side in the robot's field of view is further from one camera than the other. This results in a longer 'shadow', which biases the bearing measurement so as to overestimate its magnitude. Furthermore the magnitude of this bias increases with the magnitude of the bearing, meaning that during a turn, the change in bearing of landmarks is exaggerated. This leads to an overestimation in the angular velocity.

There are two reasons that could explain why it continues to estimate a slight turn while actually moving straight. The first is the measurement bias again. As the robot moves forward the landmarks pass by in the right hand side of its field of view. As it approaches and passes them, their bearing measurement increases, and with it the bias. This would falsely indicate a slight left hand turn. The second reason is that the velocity estimate could lag behind the true value as was observed when estimating the pose without odometry. It is unlikely though that this is to blame. The reason the zero-odometry pose estimate lagged was because its movement model assumes constant pose, which describes reality poorly. The velocity estimate movement model however assumes constant forward and rotational velocity, which describes the robot's motion well on the straights.

It has not been possible to ascertain why visual odometry significantly underestimated the forward velocity in the third dataset. It could be because this run was performed on bare wood rather than a rug, but it is unclear how this would have caused the error. The wood surface was slightly reflective but there is no indication that this affected feature detection, and there were no reflections on the left most leg where the forward velocity was underestimated the most.

Overall the combined estimation method was not very robust. Five datasets were recorded, and while it was possible to find a set of covariance matrices for the process and measurement noise that enabled the tracking algorithm to successfully navigate several of them, no set was found that could navigate all of them without getting lost. The algorithm typically failed when cornering where it would over or underestimate the change in heading. This is likely due to three factors. Firstly, at the corners the robot experienced the greatest acceleration and the movement model

assumed zero acceleration. Secondly, when cornering typically only one landmark was in the camera field of view, and finally the systematic bias discussed above meant that the measurement model would be looking for landmarks in the wrong place after a significant change in heading and reject correct detections as outliers. Furthermore, since there were always many erroneous measurements, when the estimate drifted too far from the true state these would be misidentified as landmarks and compound the error.

5 Conclusion

This work has demonstrated that it is possible to achieve workable tracking using solely optical camera input and a very simple EKF implementation. It also shows how feature mapping for visual odometry can be performed using the Kalman filter in a way that is similar to, but much simpler than Van Hamme et al. (2015). This simple approach, while not robust, makes this work useful from a pedagogic standpoint for exploring the methods and challenges of visual odometry.

The greatest potential for improvement lies in a different method of feature detection. The method used was known to be very crude but I wanted the focus to be on the estimation rather than computer vision methods. In retrospect, the systematic bias that was introduced made it difficult to judge the performance of the EKF tracking. It would have been a more valuable experience to use a method that did not produce such biased measurements. It would be possible to model this bias, but there exist many other simple feature detection methods that would serve much better.

A single camera approach would also have been preferred. Using two cameras for feature detection introduced all the difficulties of camera calibration without the benefits of full stereo vision. The method used for judging the range to detected features could have been performed equally well with a single camera. In these two respects (bias and calibration) this work illustrates a path that even from a pedagogic standpoint it is perhaps best not to take.

Range measurement relied on known camera height and tilt relative to the floor. This is very limiting to both the kinds of environment that can be navigated and to the robot's design. Van Hamme et al. (2015) show how this method of tracking features in the ground plane can be made more robust in the case of an autonomous road vehicle. However it is difficult to imagine how it might be applied in rough terrain, aerial vehicles or novel robot designs as discussed in section 1.

I think the most interesting way to extend this work would be to start without the map of known landmarks, focus on improving the robustness and flexibility of the visual odometry, and incorporate vSLAM to map the most reliable features. It would also be straightforward to add an acceleration estimation EKF and feed these as a control input to the second filter tracking the velocities. The measurement model would be based on the previous two frames at $k - 1$ and $k - 2$. This should make the estimation more robust to changes in velocity.

It would also be interesting to investigate how to make the visual odometry able to cope with features that move relative to the environment. This includes physically moving objects, but also

reflections, shadows from a moving light source, the robot's own shadow, etc. The method presented here using the Kalman filter to perform feature matching seems like a promising approach to be able to reject such erroneous features. It would however still be vulnerable to any movement that monopolises the detected features, for example a moving object that takes up the majority of the field of view.

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